



Features of linear relationships

Task A: The constant rate of change

1) Classify these relationships into **linear** and non-**linear**.

Linear

$$y = 4x \quad y = x + 4$$

$$4y = x \quad x + y = 4$$

$$y = \frac{1}{4}x \quad y = 0.4x$$

Exponent of 1

Non-linear

$$y = 4x^2 \quad y^4 = x + 4$$

$$y = 2x^4 \quad y = x^4 + 4$$

$$y = x^{0.4}$$

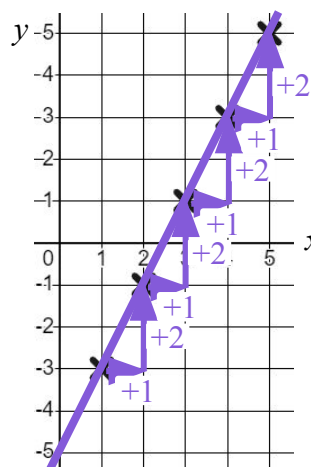
Exponents not 1

2) Complete the table for the given range of x values. Plot the relationships and show on the graph their constant rate of change.

a) $y = 2x - 5$

x	1	2	3	4	5
y	-3	-1	1	3	5

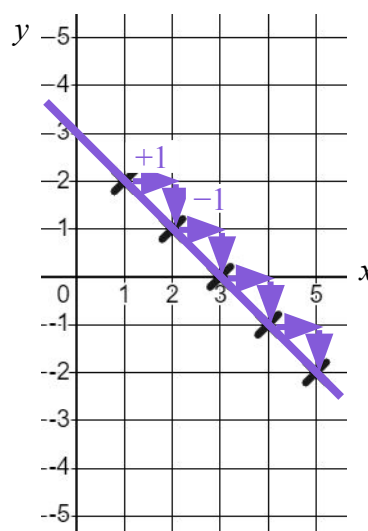
Because of the constant rate of change this is a **linear** relationship.



b) $y = 3 - x$

x	1	2	3	4	5
y	2	1	0	-1	-2

Because of the constant rate of change this is a **linear** relationship.

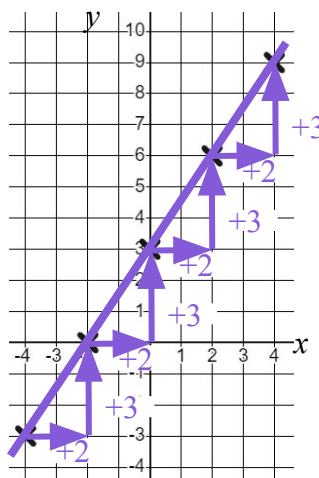


3) This set of coordinates has a linear relationship with a constant rate of change.

Find the missing values, plot the graph, and show the constant rate of change.

x	-4	-2	0	2	4
y	-3	0	3	6	9

Constant rate of change makes a **linear** relationship.



Task B: Graphical and algebraic representations

1) Match the coordinate pairs to their respective relationships and graphs.

F $y = \frac{x+2}{2}$

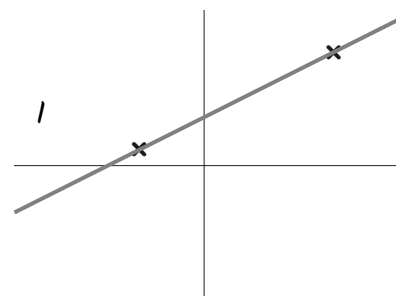
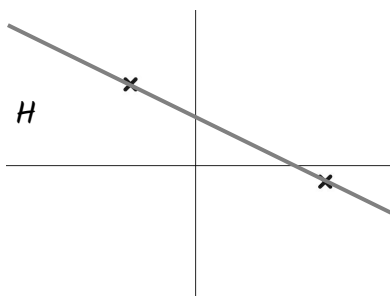
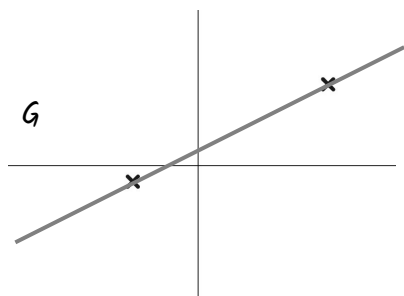
D $y = 3 - \frac{1}{2}x$

E $y = \frac{1}{2}x + 3$

C (-4, -1) and (8, 5)

B (-4, 5) and (8, -1)

A (-4, 1) and (8, 7)



Matching coordinates to quadrants is a good place to start.